Understanding Comprehensions in Python

A powerful and concise way to create collections

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What is a Comprehension?

- A **compact syntax** for creating sequences (lists, sets, dictionaries).
- Improves **readability** and **conciseness**.
- **Replaces multiple lines** of loop-based code.

Output Collection Condition

```
[x + 1 for x in range(20) if x % 2 == 0]
```

Do this for this collection in this situation

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List Comprehensions - Syntax

Syntax:

```
[expression for item in iterable if condition]
```

Example:

- **Traditional loop:**

```
squares = []
for x in range(10):
  if x % 2 == 0:
        squares.append(x**2)
```

- **Comprehension:**

```
squares = [x**2 for x in range(10) if x % 2 == 0]
```

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List Comprehensions – More Examples

```
# Convert strings to uppercase
names:list[str] = ["alice", "bob", "carol"]
uppercase_names:list[str] = [name.upper() for name in name
```

Output -> `["ALICE", "BOB", "CAROL"]`

```
# Flatten a list of lists
matrix:list[list[int]] = [[1, 2], [3, 4]]
flat:list[int] = [num for row in matrix for num in row]
```

Output -> `[1, 2, 3, 4]`

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Set and Dictionary Comprehensions

```
# Set comprehension
unique_lengths = {len(word) for word in ["hi", "hello", "hey", "hi"]}
```

Output -> `{2, 5, 3, 2}`

```
# Dictionary comprehension
Squares_dict = {x: x**2 for x in range(5)}
```

Output -> `{1: 1, 2: 4, 3: 9, 4: 16, 5: 25}`

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When Not to Use Comprehensions

- Avoid overuse: **readability first**!
- **Don't use for side-effects** (e.g., printing or modifying external state).
- **Nested comprehensions** can become **hard to read**.

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Quick Quiz

Comprehension:

```
[[x * y for y in range(1, 4)] for x in range(1, 4)]
```

Options:

A) [[1, 2, 3], [2, 4, 6], [3, 6, 9]]

B) [[1, 2, 3], [1, 4, 9], [1, 8, 27]]

C) [[1, 2, 3, 4], [2, 4, 6, 8], [3, 6, 9, 12]]

D) [(1, 1), (2, 2), (3, 3)]

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Quick Quiz – Correct Answer

Comprehension:

```
[[x * y for y in range(1, 4)] for x in range(1, 4)]
```

Options:

✅ A) `[[1, 2, 3], [2, 4, 6], [3, 6, 9]]`

❌ B) `[[1, 2, 3], [1, 4, 9], [1, 8, 27]]`

❌ C) `[[1, 2, 3, 4], [2, 4, 6, 8], [3, 6, 9, 12]]`

❌ D) `[(1, 1), (2, 2), (3, 3)]`

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Quick Quiz - Explanation

Comprehension:

```
[[x * y for y in range(1, 4)] for x in range(1, 4)]
```

Equivalent traditional loop:

```
result = []
for x in range(1, 4):
  inner_list = []
  for y in range(1, 4):
        inner_list.append(x * y)
  result.append(inner_list)
```

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